

Tuto 7

Consider the following relational schema:

Movie (title, DirectorName, duration, Nationality)

Showing (TheaterName, title, StartTime, Version)

Play (ActorName, title)

Product (ProducerName, title)

Seen (ViewerName, title)

The table instances are located at the end of this document. We would like to perform graph-oriented modeling of this database.

A) Graph-oriented modeling

Propose a graph-oriented model with:

- An exact definition of the nodes and their properties.
- An exact definition of the edges and their properties.
- Draw the resulting graph (or part of the graph).

B) Implementation in Neo4j

- Provide the scripts to implement this database in Neo4j, showing the insertion of three instances of each concept.
- The complete scripts should be completed as a homework assignment.

C) Querying the database using Cypher queries

Write the following queries in Cypher and verify their execution in Neo4j:

1. List all the movies Kevin Costner has seen.
2. List all the actors who have starred in films directed by Spielberg.
3. Give the titles of the films shown in the "Alpha" theater in their original language.
4. Find the films starring Tom Cruise that are shown in the "Beta" theater. 5. List the films produced by Adam Moore.
6. Find the viewers who saw films in which they also starred.
7. List the films screened at a given time (e.g., "3:00 PM").
8. Display the films in which Amanda Peet and Aaron Smith starred together.

9. Find the producers who worked on films screened in French.
10. Display the number of films seen by each viewer (including the top 3).

INSTANCES

Film Table (title, Director Name, duration, Nationality)

Titre	NomRéalisateur	Durée	Nationalité
Waterworld	Rickley Moore	250	US
Body Guard	Mark Gordon	120	US
Les Incorruptibles	Rickley Moore	160	US
Perfect World	Clint Eastwood	150	US
Le Dernier Chateau	Rod Lurie	132	US
Le Negociateur	David Auberman	140	US
Le Destin De Will Hunting	Gus Van Sant	120	US
High Crimes	Carl Francklin	115	US
Le Parrain	Francis Ford Coppola	168	US
Spy Games	Tony Scott	127	US
Nous Etions Soldat	Randall Wallace	139	US
Top Gun	Tony Scott	127	US
Men	Rickley Moore	140	US
Le Pacificateur	Ashley Jude	135	US
Il Faut Sauver Le soldat Ryan	Spielberg	162	US
ET	Spielberg	165	US
Indiana Jones	Spielberg	170	US

Session Table (RoomName, Title, StartTime, Version)

NomSalle	Titre	HeureDeb	Version
Alpha	Indiana Jones	13:00	VF
Alpha	Waterworld	15:00	VF
Alpha	Waterworld	17:00	VO

Alpha	Body Guard	21:00	VO
Beta	Waterworld	15:00	VF
Beta	Perfect World	19:00	VO
Ibn Zeidoun	Les Incorruptibles	13:00	VO
Ibn Zeidoun	ET	11:00	VO
Ibn Zeidoun	Top Gun	15:00	VO
Alpha	Top Gun	09:00	VF
Beta	Men	11:00	VF
Beta	Le Pacificateur	23:00	VF

Table Plays (ActorName, title)

NomAuteur	Titre
Kevin Costner	Waterworld
Kevin Costner	Perfect World
Kevin Costner	Les Incorruptibles
Kevin Costner	Body Guard
Clint Eastwood	Perfect World
Tom Cruise	Top Gun
Tom Cruise	Men
Nicole Kidman	Men
Nicole Kidman	Le Pacificateur
Harrison Ford	Indiana Jones
Tom Hanks	Il Faut Sauver Le soldat Ryan
Aaron Smith	Indiana Jones
Aaron Smith	Il Faut Sauver Le soldat Ryan
Aaron Smith	ET
Amanda Peet	Indiana Jones
Amanda Peet	Il Faut Sauver Le soldat Ryan
Amanda Peet	ET
Amanda Peet	Top Gun

Product Table (ProducerName, title)

NomProducteur	Titre
Clint Eastwood	Perfect World
Kevin Costner	Waterworld
Adam Moore	ET
Adam Moore	Men
Alex May	Top Gun

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Roger Taylor	Indiana Jones
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Table Vu (NomSpectateur, titre)

NomSpectateur	Titre
Kevin Costner	Waterworld
Kevin Costner	Perfect World
Kevin Costner	Les Incorruptibles
Kevin Costner	Body Guard
Kevin Costner	ET
Tom Cruise	Top Gun
Tom Cruise	Men
Tom Hanks	Men